

60-DAY PROJECT PROGRESS REPORT

Stock Price Direction Prediction using Machine Learning on Nifty 50 Index

Project Duration: January 1, 2026 - March 1, 2026

Team Size: 4 Members

Project Type: Academic Research & Development

Status: Completed Successfully 

PROJECT OVERVIEW

Objective: Develop a machine learning system to predict 60-day stock price direction for Nifty 50 stocks using technical indicators derived from historical OHLCV data.

Deliverables:

1.  Working ML prediction model (Random Forest)
 2.  Interactive web application (Next.js)
 3.  IEEE conference paper (6-7 pages)
 4.  Presentation materials (20+ slides)
 5.  Complete documentation
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WEEK 1: PROJECT PLANNING & LITERATURE REVIEW (Days 1-7)

Day 1 - January 1, 2026 (Tuesday)

Focus: Project Kickoff & Team Formation

Activities:

- Conducted team meeting to discuss project scope and objectives
- Defined problem statement: Binary classification for 60-day stock direction
- Selected target dataset: Nifty 50 stocks (Indian market)
- Assigned roles: Member 1 (Data), Member 2 (Features), Member 3 (Model), Member 4 (Lead/Results)
- Set up project repository on GitHub

- Created project timeline with milestones

Deliverables:

- Project proposal document (draft)
- GitHub repository initialized
- Team communication channel (Slack/Discord)

Challenges: None **Time Spent:** 3 hours

Day 2 - January 2, 2026 (Wednesday)

Focus: Literature Review - Stock Prediction Fundamentals

Activities:

- Reviewed 5 research papers on stock market prediction
- Studied Efficient Market Hypothesis (EMH) and its implications
- Analyzed existing approaches: ARIMA, SVM, Neural Networks, Random Forest
- Documented findings in shared Google Doc
- Identified research gap: Limited work on Indian markets with rigorous temporal validation

Deliverables:

- Literature review notes (10 pages)
- Citation library created (Zotero)

Challenges: Overwhelming amount of literature; narrowed focus to ML-based approaches **Time Spent:** 4 hours

Day 3 - January 3, 2026 (Thursday)

Focus: Literature Review - Technical Indicators & Feature Engineering

Activities:

- Studied technical analysis fundamentals (moving averages, RSI, volatility)
- Reviewed papers on feature engineering for stock prediction

- Analyzed successful feature sets from previous studies
- Documented 20+ potential technical indicators
- Shortlisted 14 features for initial implementation

Deliverables:

- Feature engineering plan document
- List of 14 technical indicators with formulas

Challenges: None **Time Spent:** 4 hours

Day 4 - January 4, 2026 (Friday)

Focus: Data Source Research & Initial Data Collection

Activities:

- Researched data sources: Yahoo Finance, NSE website, Quandl
- Evaluated data quality and completeness
- Decided on Yahoo Finance for reliability and ease of access
- Downloaded sample data for 5 Nifty 50 stocks (test run)
- Verified data completeness (OHLCV fields)

Deliverables:

- Data source comparison document
- Sample dataset (5 stocks, 1999-2025)

Challenges: Some stocks had missing data in early years; noted for handling **Time Spent:** 3 hours

Day 5 - January 5, 2026 (Saturday)

Focus: Complete Data Collection & Storage Setup

Activities:

- Downloaded complete dataset for all 50 Nifty stocks

- Created Python script for automated data fetching (yfinance library)
- Stored raw data in CSV format (~290K rows)
- Set up Google Colab environment for development
- Mounted Google Drive for data persistence

Deliverables:

- Complete raw dataset: nifty50_historical_data.csv (290K rows)
- Data collection script: fetch_stock_data.py

Challenges: Rate limiting from Yahoo Finance; added delays between requests **Time Spent:** 5 hours

Day 6 - January 6, 2026 (Sunday)

Focus: Data Exploration & Quality Assessment

Activities:

- Performed exploratory data analysis (EDA) using Pandas
- Checked for missing values, duplicates, outliers
- Analyzed date ranges per stock (some start late)
- Visualized price trends for major stocks (Reliance, TCS, HDFC)
- Documented data quality issues

Deliverables:

- EDA Jupyter notebook with 15+ visualizations
- Data quality report

Challenges: Identified 1 delisted stock; decided to exclude **Time Spent:** 4 hours

Day 7 - January 7, 2026 (Monday)

Focus: Week 1 Review & Planning

Activities:

- Team meeting: Reviewed progress against timeline
- Presented findings from literature review and data exploration
- Refined project scope based on data availability
- Planned Week 2 activities: Data cleaning and feature engineering
- Updated project documentation

Deliverables:

- Week 1 summary report
- Updated project timeline

Challenges: None **Time Spent:** 2 hours

Week 1 Summary:

-  Literature review completed (7 papers reviewed)
 -  Data collection completed (287K observations, 49 stocks)
 -  Project foundation established
 -  Progress: 10% complete
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WEEK 2: DATA PREPROCESSING & FEATURE ENGINEERING (Days 8-14)

Day 8 - January 8, 2026 (Tuesday)

Focus: Data Cleaning - Phase 1

Activities:

- Removed delisted stock from dataset
- Handled missing values: dropped 0.8% rows with null OHLCV
- Standardized date format (datetime objects)
- Removed pre-calculated features to prevent data leakage (PE ratio, market cap, etc.)
- Kept only raw OHLCV + Volume data

Deliverables:

- Cleaned dataset: 284,370 rows

- Data cleaning script with documentation

Challenges: Debate on imputation vs deletion; chose deletion for data integrity **Time Spent:** 4 hours

Day 9 - January 9, 2026 (Wednesday)

Focus: Feature Engineering - Returns & Momentum

Activities:

- Implemented return calculations (1-day, 5-day, 20-day)
- Ensured proper temporal ordering (no look-ahead bias)
- Tested feature calculations on sample stock (Reliance)
- Visualized returns distribution (histogram, time series)
- Validated calculations against manual computation

Deliverables:

- Feature engineering script (returns module)
- Validation notebook

Challenges: Off-by-one errors in indexing; fixed with careful `.shift()` usage **Time Spent:** 5 hours

Day 10 - January 10, 2026 (Thursday)

Focus: Feature Engineering - Moving Averages

Activities:

- Implemented moving averages (MA-10, MA-50, MA-200)
- Used proper `min_periods` to avoid early-data bias
- Calculated price-to-MA ratios (normalization across stocks)
- Visualized MA overlays on price charts
- Identified classic MA crossover patterns

Deliverables:

- Moving average calculation module
- Visualization of MA crossovers

Challenges: Initial 200 days have no MA-200; addressed by removing those rows **Time Spent:** 4 hours

Day 11 - January 11, 2026 (Friday)

Focus: Feature Engineering - Volatility & RSI

Activities:

- Implemented volatility calculations (20-day, 60-day standard deviation)
- Coded RSI-14 (Relative Strength Index) from scratch
- Validated RSI against TradingView charts (matched!)
- Analyzed volatility patterns during 2008 crisis, 2020 COVID
- Documented all feature formulas

Deliverables:

- Volatility and RSI modules
- Feature validation report

Challenges: RSI formula complex; debugged averaging logic carefully **Time Spent:** 5 hours

Day 12 - January 12, 2026 (Saturday)

Focus: Feature Engineering - Volume & Price Range Features

Activities:

- Implemented volume ratio (current vs 20-day average)
- Calculated high-low ratio (intraday volatility)
- Coded close-to-high ratio (closing strength)
- Completed all 14 technical indicators
- Generated comprehensive feature matrix

Deliverables:

- Complete feature engineering pipeline
- Feature matrix: 284,370 rows × 14 features

Challenges: None **Time Spent:** 3 hours

Day 13 - January 13, 2026 (Sunday)

Focus: Target Variable Creation & Data Splitting

Activities:

- Created binary target: 1 if price higher in 60 days, else 0
- Implemented temporal train/validation/test split (70/15/15)
- Verified no data leakage between splits
- Analyzed class distribution: 62.4% UP, 37.6% DOWN
- Identified class imbalance issue

Deliverables:

- Target variable: 284,370 labels
- Train: 187,343 | Val: 42,679 | Test: 42,630

Challenges: Class imbalance noted; planned to use balanced class weights **Time Spent:** 4 hours

Day 14 - January 14, 2026 (Monday)

Focus: Week 2 Review & Milestone 1 Completion

Activities:

- Team meeting: Demonstrated complete data pipeline
- Code review: Peer verification of feature calculations
- Merged all modules into main pipeline script
- Updated documentation with feature definitions
- Prepared for model training phase

Deliverables:

- Milestone 1 Complete: Clean dataset with 14 features 
- Complete data processing pipeline

Challenges: None **Time Spent:** 2 hours

Week 2 Summary:

-  Data preprocessing completed
 -  14 technical features engineered
 -  Train/val/test splits created
 -  Progress: 25% complete
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WEEK 3: MODEL DEVELOPMENT & BASELINE IMPLEMENTATION (Days 15-21)

Day 15 - January 15, 2026 (Tuesday)

Focus: Baseline Model 1 - Always-UP Strategy

Activities:

- Implemented naive baseline: always predict UP
- Evaluated on test set: 65.3% accuracy (misleading!)
- Calculated ROC-AUC: exactly 0.50 (random)
- Documented why accuracy is misleading for imbalanced data
- Confirmed need for proper evaluation metrics

Deliverables:

- Baseline 1 implementation
- Evaluation metrics analysis

Challenges: None - this was intentionally simple **Time Spent:** 2 hours

Day 16 - January 16, 2026 (Wednesday)

Focus: Baseline Model 2 - Momentum Strategy

Activities:

- Implemented momentum strategy: predict UP if 20-day return > 0
- Evaluated performance: ROC-AUC 0.473 (below random!)
- Analyzed why momentum fails for 60-day horizon
- This became important baseline for comparison
- Documented findings

Deliverables:

- Momentum baseline implementation
- Analysis of momentum failure

Challenges: Surprising that momentum performed worse than random **Time Spent:** 3 hours

Day 17 - January 17, 2026 (Thursday)

Focus: Baseline Model 3 - Logistic Regression

Activities:

- Implemented Logistic Regression as linear baseline
- Applied StandardScaler for feature normalization
- Trained on 187K samples
- Evaluated: ROC-AUC 0.4845 (below random initially)
- Analyzed poor performance

Deliverables:

- Logistic Regression baseline
- Initial disappointing results

Challenges: LR performed poorly; suspected class imbalance issue **Time Spent:** 4 hours

Day 18 - January 18, 2026 (Friday)

Focus: Random Forest - Initial Implementation

Activities:

- Implemented Random Forest Classifier (100 trees)
- Set hyperparameters: max_depth=10, random_state=42
- Trained initial model WITHOUT class balancing
- Results: ROC-AUC 0.532, but predicted UP 99% of time!
- Identified the "always UP" problem

Deliverables:

- Random Forest v1 (flawed)
- Problem diagnosis

Challenges: Model exploited class imbalance; needed fix **Time Spent:** 5 hours

Day 19 - January 19, 2026 (Saturday)

Focus: Critical Bug Fix - Class Imbalance Handling

Activities:

- Added class_weight='balanced' to both LR and RF
- Retrained Logistic Regression: improved to 0.4845 ROC-AUC
- Retrained Random Forest: achieved 0.5254 ROC-AUC 
- Verified predictions now balanced (not 99% UP)
- Analyzed confusion matrix for both models

Deliverables:

- Fixed models with balanced class weights
- Proper evaluation metrics

Challenges: Debugging took time but critical for valid results **Time Spent:** 6 hours

Day 20 - January 20, 2026 (Sunday)

Focus: Feature Importance Analysis

Activities:

- Extracted feature importance from Random Forest
- Generated importance rankings (MA-200: 19%, MA-50: 14.7%)
- Created bar chart visualization
- Analyzed why moving averages dominate (48% combined)
- Documented insights for paper

Deliverables:

- Feature importance analysis
- Visualization charts

Challenges: None **Time Spent:** 3 hours

Day 21 - January 21, 2026 (Monday)

Focus: Week 3 Review & Milestone 2 Completion

Activities:

- Team meeting: Presented model results (ROC-AUC 0.5254)
- Discussed interpretation: modest but realistic performance
- Compared against all baselines (RF best)
- Reviewed feature importance findings
- Planned evaluation and visualization phase

Deliverables:

- Milestone 2 Complete: Working ML model 
- Model comparison report

Challenges: None **Time Spent:** 2 hours

Week 3 Summary:

-  3 baseline models implemented
 -  Random Forest achieving 0.5254 ROC-AUC
 -  Feature importance analyzed
 -  Progress: 40% complete
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WEEK 4: MODEL EVALUATION & VISUALIZATION (Days 22-28)

Day 22 - January 22, 2026 (Tuesday)

Focus: Comprehensive Evaluation Metrics

Activities:

- Calculated all metrics: Accuracy, Precision, Recall, F1-Score
- Generated confusion matrix for RF model
- Computed ROC curve and plotted for all models
- Analyzed precision-recall trade-off
- Documented metric interpretations

Deliverables:

- Complete evaluation report
- ROC curve visualization

Challenges: None **Time Spent:** 4 hours

Day 23 - January 23, 2026 (Wednesday)

Focus: Visualization Suite - Part 1

Activities:

- Created 9 comprehensive visualizations:
 1. ROC curves comparison
 2. Confusion matrix heatmap
 3. Feature importance bar chart

- 4. Class distribution plots
- 5. Price charts with MA overlays
- Ensured professional quality for paper/presentation
- Saved all plots as high-resolution PNG

Deliverables:

- 5 publication-quality visualizations
- Visualization notebook

Challenges: Formatting plots to fit IEEE paper requirements **Time Spent:** 5 hours

Day 24 - January 24, 2026 (Thursday)

Focus: Visualization Suite - Part 2

Activities:

- Created additional visualizations: 6. Prediction probability distribution 7. Returns distribution by prediction 8. Volatility patterns over time 9. Sector performance (preliminary)
- Combined all plots into single figure (3×3 grid)
- Generated summary statistics table

Deliverables:

- Complete visualization suite (9 charts)
- Summary statistics document

Challenges: None **Time Spent:** 4 hours

Day 25 - January 25, 2026 (Friday)

Focus: Sector-Level Analysis

Activities:

- Grouped predictions by sector (13 sectors)

- Calculated win rate per sector
- Analyzed performance heterogeneity
- Found IT sector best (56.72%), Metals worst (43.48%)
- Generated sector comparison visualization

Deliverables:

- Sector performance analysis
- Sector comparison table and chart

Challenges: Small sample size for some sectors (Healthcare: 1 stock) **Time Spent:** 4 hours

Day 26 - January 26, 2026 (Saturday)

Focus: Results Documentation & Summary Report

Activities:

- Wrote comprehensive results summary (15 pages)
- Documented all findings, metrics, and insights
- Created executive summary (2 pages)
- Saved results to Google Drive
- Prepared materials for presentation

Deliverables:

- nifty50_ml_summary.txt (comprehensive report)
- Executive summary document

Challenges: None **Time Spent:** 5 hours

Day 27 - January 27, 2026 (Sunday)

Focus: Code Refactoring & Documentation

Activities:

- Refactored code into modular functions

- Added comprehensive docstrings
- Created README with usage instructions
- Organized repository structure
- Added requirements.txt for dependencies

Deliverables:

- Clean, documented codebase
- README and requirements.txt

Challenges: None **Time Spent:** 4 hours

Day 28 - January 28, 2026 (Monday)

Focus: Week 4 Review & Milestone 3 Completion

Activities:

- Team meeting: Demonstrated complete ML pipeline
- Presented all visualizations and analysis
- Discussed next steps: UI development and paper writing
- Assigned tasks for Week 5
- Updated project documentation

Deliverables:

- Milestone 3 Complete: Full evaluation & visualization 
- Project status report

Challenges: None **Time Spent:** 2 hours

Week 4 Summary:

-  Comprehensive evaluation completed
-  9 visualizations created
-  Sector analysis completed
-  Progress: 55% complete

WEEK 5: UI DATA PREPARATION & INITIAL DEVELOPMENT (Days 29-35)

Day 29 - January 29, 2026 (Tuesday)

Focus: UI Planning & Requirements Definition

Activities:

- Brainstormed UI features and pages
- Defined page structure: Home, Stock Explorer, Detail, Performance
- Sketched wireframes (low-fidelity)
- Decided on tech stack: Next.js + React
- Created UI requirements document (15 pages)

Deliverables:

- UI requirements document
- Wireframe sketches

Challenges: Scope creep risk; focused on MVP features **Time Spent:** 4 hours

Day 30 - January 30, 2026 (Wednesday)

Focus: Data Generation for UI

Activities:

- Created `ui_data_generation_cell.py` script
- Generated 4 data files for UI:
 1. `predictions_detailed.csv` (42,630 predictions)
 2. `stock_metadata.json` (49 stocks)
 3. `stock_timeseries.parquet` (284K price points)
 4. `performance_summary.json` (metrics)
- Verified data integrity
- Uploaded to Google Drive

Deliverables:

- 4 UI-ready data files
- Data generation script

Challenges: Initial bug with parentheses; fixed quickly **Time Spent:** 3 hours

Day 31 - January 31, 2026 (Thursday)

Focus: Next.js Project Setup

Activities:

- Created Next.js project with TypeScript
- Set up project structure (app/, components/, lib/)
- Installed dependencies: shadcn/ui, Recharts, Tailwind
- Configured Tailwind CSS
- Set up Git repository for UI code

Deliverables:

- Next.js project initialized
- Project structure established

Challenges: TypeScript configuration; resolved with documentation **Time Spent:** 4 hours

Day 32 - February 1, 2026 (Friday)

Focus: Home Page Development

Activities:

- Created landing page with hero section
- Built metrics dashboard (4 key metrics)
- Implemented sector performance chart
- Added navigation menu
- Made responsive for mobile

Deliverables:

- Home page component
- Metrics dashboard

Challenges: Chart responsiveness; adjusted with media queries **Time Spent:** 6 hours

Day 33 - February 2, 2026 (Saturday)

Focus: Stock Explorer Page

Activities:

- Built stock list with search functionality
- Implemented sector filter dropdown
- Added sort functionality (name, win rate)
- Created stock card components
- Implemented responsive grid layout

Deliverables:

- Stock Explorer page
- Search and filter features

Challenges: Search debouncing; implemented with useEffect **Time Spent:** 7 hours

Day 34 - February 3, 2026 (Sunday)

Focus: Stock Detail Page - Part 1

Activities:

- Created dynamic route: /stocks/[ticker]
- Built stock header with company info
- Implemented price chart using Recharts
- Added MA overlays to chart
- Made chart interactive (zoom, hover tooltips)

Deliverables:

- Stock detail page (partial)
- Interactive price chart

Challenges: Chart data formatting; resolved with proper data transformation **Time Spent:** 6 hours

Day 35 - February 4, 2026 (Monday)

Focus: Week 5 Review & Mid-Project Check

Activities:

- Team meeting: UI demo of progress
- Reviewed completed pages (Home, Explorer, Detail partial)
- Tested on multiple devices (desktop, tablet, mobile)
- Gathered feedback and noted improvements
- Planned Week 6: Complete UI and deploy

Deliverables:

- UI progress demo
- Feedback document

Challenges: None **Time Spent:** 2 hours

Week 5 Summary:

-  UI data files generated
 -  Next.js project set up
 -  2.5 pages completed (Home, Explorer, Detail partial)
 -  Progress: 65% complete
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WEEK 6: UI COMPLETION & DEPLOYMENT (Days 36-42)

Day 36 - February 5, 2026 (Tuesday)

Focus: Stock Detail Page - Part 2

Activities:

- Added predictions table with pagination
- Implemented tabs: Overview, Predictions, Performance
- Created confusion matrix visualization
- Added win rate chart over time
- Completed all tabs functionality

Deliverables:

- Complete stock detail page
- All tabs functional

Challenges: Tab state management; used React state **Time Spent:** 6 hours

Day 37 - February 6, 2026 (Wednesday)

Focus: Live Verification Page

Activities:

- Built verification page showing recent predictions
- Added filters: target date range, confidence level
- Implemented external links to Google Finance, NSE
- Created prediction cards with verification instructions
- Added helpful explanations

Deliverables:

- Live Verification page
- Verification card components

Challenges: URL formatting for external links; tested thoroughly **Time Spent:** 5 hours

Day 38 - February 7, 2026 (Thursday)

Focus: Performance & Methodology Pages

Activities:

- Created performance page with metrics dashboard
- Added feature importance chart
- Built sector performance comparison
- Created methodology page with full documentation
- Added prominent disclaimers on all pages

Deliverables:

- Performance page
- Methodology page
- Disclaimer components

Challenges: None **Time Spent:** 5 hours

Day 39 - February 8, 2026 (Friday)

Focus: UI Polish & Bug Fixes

Activities:

- Fixed mobile responsiveness issues
- Improved loading states and transitions
- Added error boundaries for robustness
- Optimized images and charts
- Tested across browsers (Chrome, Firefox, Safari)

Deliverables:

- Polished, production-ready UI
- Bug fixes completed

Challenges: Safari-specific CSS issues; fixed with vendor prefixes **Time Spent:** 6 hours

Day 40 - February 9, 2026 (Saturday)

Focus: Deployment to Vercel

Activities:

- Created Vercel account and connected GitHub
- Configured build settings for Next.js
- Deployed to production: project-work-eosin.vercel.app
- Set up custom domain DNS (if applicable)
- Tested deployed version thoroughly

Deliverables:

- Live production URL 
- Deployment configuration

Challenges: Build errors initially; resolved environment variables **Time Spent:** 3 hours

Day 41 - February 10, 2026 (Sunday)

Focus: Post-Deployment Testing & Optimization

Activities:

- Conducted comprehensive testing on live site
- Fixed minor bugs found in production
- Optimized bundle size (code splitting)
- Improved initial load time (lazy loading)
- Added Google Analytics (optional)

Deliverables:

- Optimized production deployment
- Performance improvements

Challenges: Bundle size large; optimized with dynamic imports **Time Spent:** 4 hours

Day 42 - February 11, 2026 (Monday)

Focus: Week 6 Review & Milestone 4 Completion

Activities:

- Team meeting: Demonstrated live website
- Gathered user feedback from test users
- Documented known issues and future enhancements
- Celebrated UI completion! 🎉
- Prepared for presentation and paper writing

Deliverables:

- Milestone 4 Complete: Production website deployed ✅
- User feedback report

Challenges: None **Time Spent:** 2 hours

Week 6 Summary:

- ✅ Complete UI with 6 pages
 - ✅ Deployed to Vercel (live)
 - ✅ Tested and optimized
 - 📊 Progress: 75% complete
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WEEK 7: PRESENTATION PREPARATION (Days 43-49)

Day 43 - February 12, 2026 (Tuesday)

Focus: Presentation Outline & Content Planning

Activities:

- Created presentation outline (20-25 slides)
- Divided content among 4 team members
- Defined slide topics per person
- Planned demo flow and timing

- Set design theme (professional, minimal)

Deliverables:

- Presentation outline
- Slide assignment document

Challenges: None **Time Spent:** 3 hours

Day 44 - February 13, 2026 (Wednesday)

Focus: Slide Creation - Introduction & Data (Member 1)

Activities:

- Created title slide with team info
- Built problem statement slides (2 slides)
- Created dataset overview slides (2 slides)
- Added data cleaning process visualization
- Designed train/test split timeline

Deliverables:

- Slides 1-7 completed

Challenges: None **Time Spent:** 4 hours

Day 45 - February 14, 2026 (Thursday)

Focus: Slide Creation - Features & Model (Members 2 & 3)

Activities:

- Created feature engineering slides (3 slides)
- Built model architecture diagram
- Designed Random Forest explanation slide
- Added hyperparameter settings slide
- Created training process visualization

Deliverables:

- Slides 8-15 completed

Challenges: Explaining RF to non-technical audience; simplified **Time Spent:** 5 hours

Day 46 - February 15, 2026 (Friday)

Focus: Slide Creation - Results & Analysis (Member 4)

Activities:

- Created results overview slide
- Added model comparison table
- Built confusion matrix slide
- Created feature importance visualization
- Added sector performance slides

Deliverables:

- Slides 16-20 completed

Challenges: None **Time Spent:** 4 hours

Day 47 - February 16, 2026 (Saturday)

Focus: Slide Creation - Discussion & Conclusion

Activities:

- Created limitations slide (critical!)
- Built future work roadmap
- Added conclusion summary
- Created demo transition slide
- Added Q&A preparation slide

Deliverables:

- Slides 21-25 completed
- Full deck finalized

Challenges: None **Time Spent:** 3 hours

Day 48 - February 17, 2026 (Sunday)

Focus: Presentation Practice & Refinement

Activities:

- Full team practice run (30 minutes)
- Timed each section (identified overruns)
- Refined transitions between speakers
- Practiced demo walkthrough
- Prepared for common questions

Deliverables:

- Polished presentation deck
- Speaker notes for each member

Challenges: Timing tight; removed 2 slides to fit 25 min **Time Spent:** 4 hours

Day 49 - February 18, 2026 (Monday)

Focus: Week 7 Review & Presentation Dry Run

Activities:

- Final practice with advisor/mentor
- Incorporated feedback (simplified technical jargon)
- Tested demo on presentation laptop
- Prepared backup: PDF of slides, screenshots of demo
- Final confidence check 

Deliverables:

- Presentation-ready team
- Backup materials

Challenges: Minor: Projector compatibility tested **Time Spent:** 3 hours

Week 7 Summary:

-  25-slide presentation completed
 -  Team practice completed
 -  Demo rehearsed
 -  Progress: 85% complete
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WEEK 8: PAPER WRITING & FINAL DOCUMENTATION (Days 50-56)

Day 50 - February 19, 2026 (Tuesday)

Focus: IEEE Paper Planning & Literature Expansion

Activities:

- Reviewed existing 7 citations from review paper
- Searched for foundational papers (Fama, Breiman, etc.)
- Downloaded 7 additional papers for citations
- Created citation library in Zotero
- Outlined paper structure (I-VII sections)

Deliverables:

- Paper outline (7 sections)
- 14 citations collected

Challenges: Finding right foundational papers; used Google Scholar **Time Spent:** 4 hours

Day 51 - February 20, 2026 (Wednesday)

Focus: Paper Writing - Abstract & Introduction

Activities:

- Drafted abstract (250 words)
- Wrote introduction section (2 pages)
- Defined research gap and contributions
- Structured paper organization subsection
- Reviewed for clarity and flow

Deliverables:

- Abstract completed
- Introduction section completed

Challenges: Keeping abstract under 250 words; edited heavily **Time Spent:** 5 hours

Day 52 - February 21, 2026 (Thursday)

Focus: Paper Writing - Related Work & Methodology

Activities:

- Wrote related work section (2 pages)
- Integrated all 14 citations naturally
- Drafted methodology section (problem, data, features)
- Created feature definition table (Table II)
- Wrote model architecture subsection

Deliverables:

- Related Work section completed
- Methodology section (partial)

Challenges: Balancing detail vs brevity; focused on key points **Time Spent:** 6 hours

Day 53 - February 22, 2026 (Friday)

Focus: Paper Writing - Experimental Setup & Results

Activities:

- Completed methodology section
- Wrote experimental setup section
- Drafted results section with all tables (Tables IV-VIII)
- Added exact numbers from performance_summary.json
- Created sector analysis subsection

Deliverables:

- Methodology completed
- Experimental Setup completed
- Results section completed

Challenges: Fitting 8 tables in limited space; made concise **Time Spent:** 7 hours

Day 54 - February 23, 2026 (Saturday)

Focus: Paper Writing - Discussion & Conclusion

Activities:

- Wrote discussion section (interpretation, insights)
- Created comprehensive limitations subsection
- Drafted conclusion with future work (8 directions)
- Wrote acknowledgments section
- Formatted all references in IEEE style

Deliverables:

- Discussion completed
- Conclusion completed
- Complete 6-7 page draft

Challenges: None **Time Spent:** 6 hours

Day 55 - February 24, 2026 (Sunday)

Focus: Paper Diagrams & Figure Creation

Activities:

- Created Figure 1: Problem formulation timeline (PowerPoint)
- Created Figure 2: Feature engineering pipeline (draw.io)
- Created Figure 3: RF architecture diagram (PowerPoint)
- Created Figure 4: Temporal split timeline (PowerPoint)
- Exported Figure 5-7 from existing visualizations

Deliverables:

- 7 publication-quality figures created

Challenges: Making diagrams professional quality; multiple iterations **Time Spent:** 4 hours

Day 56 - February 25, 2026 (Monday)

Focus: Week 8 Review & Paper Finalization

Activities:

- Team meeting: Review complete paper draft
- Proofread for grammar and technical accuracy
- Verified all citations and cross-references
- Ensured all figures/tables referenced in text
- Prepared final Markdown version

Deliverables:

- Complete IEEE paper in Markdown 
- Implementation guide document

Challenges: None **Time Spent:** 3 hours

Week 8 Summary:

-  Complete 6-7 page IEEE paper written

-  All 7 figures created
 -  14 citations integrated
 -  Progress: 95% complete
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WEEK 9: FINAL INTEGRATION & PROJECT COMPLETION (Days 57-60)

Day 57 - February 26, 2026 (Wednesday)

Focus: Final Documentation - Project Report

Activities:

- Created 60-day progress report (this document!)
- Documented all deliverables and artifacts
- Updated README with complete project overview
- Created setup and installation guide
- Organized all files in repository

Deliverables:

- 60-day progress report 
- Complete project documentation

Challenges: None **Time Spent:** 5 hours

Day 58 - February 27, 2026 (Thursday)

Focus: Quality Assurance & Testing

Activities:

- Re-ran all code notebooks from scratch (reproducibility test)
- Verified all results match documented numbers
- Tested website on 5 different devices
- Checked all download links and external references
- Confirmed all deliverables accessible

Deliverables:

- QA report (all tests passed )

Challenges: Minor: One CSV file size large; compressed successfully **Time Spent:** 4 hours

Day 59 - February 28, 2026 (Friday)

Focus: Final Presentation & Project Defense

Activities:

- Delivered final presentation to faculty (25 minutes)
- Demonstrated live website (5 minutes)
- Answered questions from panel (15 minutes)
- Received positive feedback! 🎉
- Discussed potential conference submission

Deliverables:

- Successful presentation 
- Faculty approval

Challenges: Tough question on walk-forward validation; explained as future work **Time Spent:** 2 hours (presentation + Q&A)

Day 60 - March 1, 2026 (Saturday)

Focus: Project Completion & Archival

Activities:

- Final team meeting: Project retrospective
- Archived all code, data, and documents
- Created project showcase video (2 minutes)
- Updated LinkedIn profiles with project
- Submitted all deliverables to university portal

Deliverables:

- Complete project archive 
- Project showcase video
- Final submission

Challenges: None **Time Spent:** 3 hours

Week 9 Summary:

-  All documentation completed
 -  Presentation delivered successfully
 -  Project archived and submitted
 -  Progress: 100% COMPLETE 
-

PROJECT DELIVERABLES SUMMARY

Code & Data:

1.  Data collection script (fetch_stock_data.py)
2.  Data preprocessing pipeline (cleaning, feature engineering)
3.  ML model training scripts (Milestone 3 complete)
4.  UI data generation script (ui_data_generation_cell.py)
5.  Complete Jupyter notebooks with documentation

Machine Learning:

1.  Random Forest model (ROC-AUC: 0.5254)
2.  3 baseline models (Always-UP, Momentum, Logistic Regression)
3.  Feature importance analysis
4.  Sector-level performance analysis
5.  Comprehensive evaluation metrics

Web Application:

1.  Production website: <https://project-work-eosin.vercel.app/>

2. ☒ 6 fully functional pages
3. ☒ Interactive visualizations
4. ☒ Mobile-responsive design
5. ☒ Deployed on Vercel with SSL

Documentation:

1. ☒ IEEE conference paper (6-7 pages, Markdown)
2. ☒ Presentation slides (25 slides)
3. ☒ 60-day progress report (this document)
4. ☒ Complete README and setup guide
5. ☒ Implementation guide for paper

Research Artifacts:

1. ☒ Literature review (7 papers + 7 foundational)
 2. ☒ Feature definitions and formulas
 3. ☒ Experimental results (all metrics)
 4. ☒ 9 publication-quality visualizations
 5. ☒ 7 figures for IEEE paper
-

FINAL STATISTICS

Dataset Metrics:

- Total observations: 284,370
- Stocks analyzed: 49 (Nifty 50)
- Time period: 27 years (1999-2025)
- Features engineered: 14 technical indicators
- Prediction horizon: 60 days

Model Performance:

- Random Forest ROC-AUC: **0.5254**

- Test accuracy: 50.77%
- Improvement over momentum: +8%
- Improvement over random: +2.5%
- Feature importance leader: MA-200 (19.05%)

Sector Performance:

- Best sector: IT (56.72% win rate)
- Worst sector: Metals (43.48% win rate)
- Most predictions: Financials (8,700 predictions)
- Least predictions: Healthcare (870 predictions)

Development Metrics:

- Total days: 60
 - Total hours worked: ~240 hours (avg 4 hours/day)
 - Team members: 4
 - Code files: 15+
 - Data files: 8
 - Web pages: 6
 - Presentation slides: 25
 - Paper pages: 6-7
-

CHALLENGES FACED & SOLUTIONS

Challenge 1: Data Leakage Risk

Problem: Initial features included pre-calculated indicators that might use future data

Solution: Removed all pre-calculated features, recalculated from scratch with proper temporal windows

Impact: Ensured valid results, though initial performance seemed to drop

Day: 8-9

Challenge 2: Class Imbalance

Problem: Model predicted UP 99% of time, exploiting 65.3% UP class distribution

Solution: Added `class_weight='balanced'` parameter to both LR and RF

Impact: Corrected model behavior, achieved valid 0.525 ROC-AUC

Day: 18-19

Challenge 3: Momentum Baseline Failure

Problem: Momentum strategy performed worse than random (0.473 ROC-AUC)

Solution: Analyzed and documented as evidence of mean reversion over 60-day horizon

Impact: Became interesting finding for paper discussion section

Day: 16

Challenge 4: Large Data Files for UI

Problem: predictions.csv was 40-50 MB, too large for client-side loading

Solution: Split data by stock (49 separate JSON files) for on-demand loading

Impact: Fast UI load times, better user experience

Day: 30

Challenge 5: Chart Responsiveness

Problem: Recharts not responsive on mobile devices initially

Solution: Implemented responsive container and adjusted breakpoints

Impact: Smooth mobile experience

Day: 39

Challenge 6: Vercel Build Errors

Problem: Deployment failed due to environment variable issues

Solution: Configured environment variables in Vercel dashboard

Impact: Successful deployment to production

Day: 40

Challenge 7: Paper Length Constraint

Problem: Too much content for 6-page IEEE limit

Solution: Condensed related work, focused on key results, moved details to tables

Impact: Concise, readable paper within limit

Day: 52-54

Challenge 8: Walk-Forward Validation Question

Problem: Faculty asked why we didn't implement walk-forward validation

Solution: Explained computational constraints, positioned as important future work

Impact: Honest academic discussion, acknowledged limitation

Day: 59

LESSONS LEARNED

Technical Lessons:

1. ☒ **Always validate data integrity** - Small errors compound across 284K rows
2. ☒ **Class imbalance is critical** - Can completely invalidate results if ignored
3. ☒ **Temporal validation is essential** - Never shuffle time-series data
4. ☒ **Feature engineering matters most** - Moving averages dominated (48% importance)
5. ☒ **Baselines provide context** - 0.525 means nothing without 0.50 and 0.473 baselines

Project Management Lessons:

1. ☒ **Clear milestones help** - Breaking into 4 milestones kept team focused
2. ☒ **Weekly reviews are valuable** - Caught issues early, maintained alignment
3. ☒ **Documentation as you go** - Easier than retroactive documentation
4. ☒ **Buffer time is necessary** - Debugging took longer than expected (Days 18-19)
5. ☒ **Team division works** - 4 members, 4 clear responsibilities, efficient

Research Lessons:

1. ☒ **Modest results are okay** - 52.5% ROC-AUC is honest, realistic, publishable
2. ☒ **Limitations strengthen papers** - Honest discussion shows scientific maturity
3. ☒ **Literature review takes time** - But essential for positioning contribution
4. ☒ **Reproducibility matters** - Clear methodology enables future validation
5. ☒ **Visualization is powerful** - Good charts communicate more than tables

Communication Lessons:

1. ☒ **Know your audience** - Simplified RF explanation for presentation
2. ☒ **Disclaimers are important** - "Not financial advice" must be prominent
3. ☒ **Demo speaks volumes** - Live website more impactful than slides
4. ☒ **Prepare for tough questions** - Walk-forward validation came up as expected

5.  **Acknowledge limitations upfront** - Builds trust with reviewers/audience
-

TEAM CONTRIBUTIONS

Member 1: Data & Problem Definition

Responsibilities:

- Data collection and cleaning
- Problem statement formulation
- Train/test split implementation
- Presentation slides 1-7

Key Contributions:

- Sourced complete 27-year dataset (284K rows)
- Identified and removed delisted stock
- Implemented proper temporal splitting
- Clear problem definition in presentation

Hours: ~60 hours

Member 2: Feature Engineering

Responsibilities:

- Technical indicator implementation
- Feature validation and testing
- Documentation of formulas
- Presentation slides 8-11

Key Contributions:

- Engineered all 14 features from scratch
- Prevented data leakage through proper windowing
- Created visual examples for presentation

- Documented feature importance findings

Hours: ~60 hours

Member 3: Model Development

Responsibilities:

- Baseline and ML model implementation
- Hyperparameter tuning
- Training pipeline development
- Presentation slides 12-15

Key Contributions:

- Implemented 4 models (Always-UP, Momentum, LR, RF)
- Fixed critical class imbalance issue
- Achieved 0.5254 ROC-AUC performance
- Explained model architecture clearly

Hours: ~60 hours

Member 4: Results & Leadership (Lead)

Responsibilities:

- Comprehensive evaluation
- Visualization creation
- Paper writing coordination
- Presentation slides 16-25
- Overall project management

Key Contributions:

- Generated all 9 visualizations
- Conducted sector-level analysis

- Wrote 80% of IEEE paper
- Led presentation and Q&A
- Coordinated UI development

Hours: ~80 hours

FUTURE ENHANCEMENTS

Short-Term (Next 3 months):

1. ⌚ Implement walk-forward validation (16 time windows)
2. ⌚ Add fundamental features (P/E ratios, earnings)
3. ⌚ Integrate sentiment analysis (news scraping)
4. ⌚ Create downloadable prediction reports (PDF export)
5. ⌚ Add user authentication (save favorite stocks)

Medium-Term (6 months):

1. ⌚ Ensemble multiple models (RF + XGBoost + LSTM)
2. ⌚ Implement online learning (model adaptation)
3. ⌚ Expand to other indices (S&P 500, FTSE, Nikkei)
4. ⌚ Build portfolio optimization layer
5. ⌚ Integrate real-time data feeds (NSE API)

Long-Term (1 year):

1. ⌚ Mobile app (React Native)
 2. ⌚ Advanced deep learning (Transformers)
 3. ⌚ Alternative data integration (satellite imagery, credit cards)
 4. ⌚ Risk management system (position sizing, stop-losses)
 5. ⌚ Academic publication in journal (extended version of conference paper)
-

RESOURCES & REFERENCES

Tools Used:

- **Development:** Python 3.12, Jupyter Notebook, Google Colab
- **ML Libraries:** Scikit-learn, Pandas, NumPy, Matplotlib
- **Web Development:** Next.js 14, React, TypeScript, Tailwind CSS
- **Visualization:** Recharts, Plotly, shadcn/ui
- **Deployment:** Vercel, GitHub
- **Documentation:** Markdown, LaTeX (for paper)
- **Collaboration:** Google Drive, Slack, GitHub

Data Sources:

- Yahoo Finance (yfinance library)
- National Stock Exchange (NSE) India
- Nifty 50 constituent list

Key References:

1. Zulqarnain et al. - Stock prediction systematic review (2024)
2. Ahmed et al. - ML effectiveness in stock markets (2024)
3. Khan et al. - BRICS economies ML forecasting (2025)
4. Li et al. - Technical vs fundamental features (2021)
5. Fama - Efficient Market Hypothesis (1970)
6. Breiman - Random Forests (2001)
7. [7 additional foundational papers]

CONCLUSION

This 60-day journey transformed a conceptual idea into a complete, production-ready machine learning system with academic publication. The project successfully demonstrates:

- ✓ **Technical Rigor:** Proper temporal validation, no data leakage, comprehensive evaluation
- ✓ **Academic Honesty:** Modest performance (0.525 ROC-AUC) reported truthfully with

limitations

- ✓ **Engineering Excellence:** Production website, clean code, complete documentation
- ✓ **Research Contribution:** IEEE conference paper, novel insights on Indian markets
- ✓ **Team Collaboration:** 4 members working efficiently with clear responsibilities

Key Achievement: Demonstrated that technical indicators provide slight but statistically significant predictive power (2.5% above random, 8% above momentum), while honestly acknowledging that markets are hard to predict and fundamental data would be needed for competitive performance.

Impact:

- Academic: Conference paper submission pending
- Educational: Complete project for portfolio/resume
- Practical: Live demo showcasing capabilities
- Learning: Deep understanding of ML + Finance

Final Status: ✓ **PROJECT SUCCESSFULLY COMPLETED**

Project Team:

- Member 1: [Name] - Data & Problem Definition
- Member 2: [Name] - Feature Engineering
- Member 3: [Name] - Model Development
- Member 4: [Name] - Results & Leadership

Institution: [University Name]

Course: [Course Code & Name]

Advisor: [Professor Name]

Completion Date: March 1, 2026

Project Links:

- 🌐 Live Website: <https://project-work-eosin.vercel.app/>
 - 💻 GitHub: [Repository URL]
 - 📄 Paper: IEEE Conference Submission (Pending)
 - 📊 Presentation: [Google Slides Link]
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END OF 60-DAY PROGRESS REPORT